

Inverse Correlation Similarity: Algorithm C

Construction, Theory, and Applications

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1 Introduction

Pearson correlation is one of the most widely used measures of statistical dependence in statistics, finance, econometrics, machine learning, signal processing, and the natural sciences.

Given two vectors

$$u, v \in \mathbb{R}^n,$$

the forward problem — computing the correlation coefficient

$$\rho(u, v)$$

— is elementary.

The inverse problem is substantially less studied:

Given a vector

$$u \in \mathbb{R}^n$$

and a target correlation

$$c \in [-1, 1],$$

construct a vector

$$v \in \mathbb{R}^n$$

such that

$$\rho(u, v) = c.$$

This paper presents Algorithm C, a constructive and provably correct algorithm for solving the inverse correlation problem in linear time.

The key observation is that Pearson correlation is equivalent to cosine similarity after mean centering. Consequently, the inverse correlation problem reduces to an inverse cosine similarity problem inside the mean-zero hyperplane.

Applications include:

- Synthetic financial return generation.
- Correlation-controlled Monte Carlo simulation.
- Adversarial statistical data synthesis.
- Covariance stress testing.
- Controlled synthetic dataset generation.

The paper ends with “The End”.

2 Mathematical Preliminaries

2.1 Pearson Correlation

Definition 2.1 (Pearson Correlation). *Let*

$$u, v \in \mathbb{R}^n$$

be nonconstant vectors.

Define the sample means

$$\bar{u} = \frac{1}{n} \sum_{i=1}^n u_i, \quad \bar{v} = \frac{1}{n} \sum_{i=1}^n v_i.$$

The Pearson correlation coefficient is

$$\rho(u, v) = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2} \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}.$$

2.2 Centered Vectors

Definition 2.2 (Centered Vector). *For*

$$u \in \mathbb{R}^n,$$

define the centered vector

$$u_c = u - \bar{u}\mathbf{1},$$

where

$$\mathbf{1} = (1, \dots, 1)^\top.$$

By construction,

$$\sum_{i=1}^n (u_c)_i = 0.$$

Thus

$$u_c$$

lies inside the hyperplane

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

2.3 Correlation as Cosine Similarity

Theorem 2.3 (Correlation-Cosine Equivalence). *For nonconstant vectors*

$$u, v \in \mathbb{R}^n,$$

$$\rho(u, v) = \frac{u_c^\top v_c}{|u_c||v_c|}.$$

Proof. Substituting the centered vectors into the definition of Pearson correlation yields

$$\rho(u, v) = \frac{\langle u_c, v_c \rangle}{|u_c||v_c|}.$$

This is exactly cosine similarity.

□

3 The Inverse Correlation Problem

Definition 3.1 (Inverse Correlation Problem). *Given*

$$u \in \mathbb{R}^n$$

and

$$c \in [-1, 1],$$

construct

$$v \in \mathbb{R}^n$$

such that

$$\rho(u, v) = c.$$

4 Geometric Structure

The centered vectors lie inside the hyperplane

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

This is an

$$(n-1)$$

-dimensional linear subspace of

$$\mathbb{R}^n.$$

Inside

$$H,$$

correlation becomes cosine similarity.

Thus the inverse correlation problem becomes a constrained angular synthesis problem.

5 Construction

Let

$$\hat{u} = \frac{u_c}{|u_c|}.$$

Choose a vector

$$w \in H$$

satisfying

$$w \perp \hat{u}, \quad |w| = 1.$$

Define

$$v_c = c\hat{u} + \sqrt{1-c^2}w.$$

Then

$$|v_c|^2 = c^2 + (1-c^2)1.$$

Furthermore,

$$\langle \hat{u}, v_c \rangle = c.$$

Hence

$$\rho(u, v_c) = c.$$

Finally define

$$v = v_c + \mu \mathbf{1},$$

where

$$\mu \in \mathbb{R}$$

is arbitrary.

Since Pearson correlation is invariant under additive shifts,

$$\rho(u, v) = \rho(u, v_c)c.$$

6 Algorithm C

6.1 Algorithm

Require

$$u \in \mathbb{R}^n, \quad c \in [-1, 1], \quad n \geq 2.$$

Ensure

$$v \in \mathbb{R}^n$$

such that

$$\rho(u, v) = c.$$

1. Compute

$$\bar{u} = \frac{1}{n} \sum_{i=1}^n u_i.$$

2. Center the vector:

$$u_c = u - \bar{u}\mathbf{1}.$$

3. Normalize:

$$\hat{u} = \frac{u_c}{|u_c|}.$$

4. Repeat:

(a) Sample

$$r \sim \mathcal{N}(0, I_n).$$

(b) Compute the centered vector

$$r_c = r - \bar{r}\mathbf{1}.$$

(c) Orthogonalize:

$$r_{\perp} = r_c - \langle r_c, \hat{u} \rangle \hat{u}.$$

until

$$|r_{\perp}| > \varepsilon.$$

5. Normalize:

$$w = \frac{r_{\perp}}{|r_{\perp}|}.$$

6. Construct

$$v_c = c\hat{u} + \sqrt{1 - c^2}w.$$

7. Choose arbitrary

$$\mu \in \mathbb{R}.$$

8. Define

$$v = v_c + \mu\mathbf{1}.$$

9. Return

$$v.$$

7 Theoretical Analysis

7.1 Correctness

Theorem 7.1 (Correctness of Algorithm C). *Algorithm C produces a vector*

$$v$$

satisfying

$$\rho(u, v) = c.$$

Proof. By construction,

$$w \perp \hat{u}, \quad |w| = 1.$$

Hence

$$\langle \hat{u}, v_c \rangle = \left\langle \hat{u}, c\hat{u} + \sqrt{1 - c^2}w \right\rangle = c.$$

Also,

$$|v_c|^2 = c^2 + (1 - c^2)1.$$

Therefore

$$\frac{u_c^\top v_c}{|u_c||v_c|} = c.$$

Since Pearson correlation equals cosine similarity of centered vectors,

$$\rho(u, v) = c.$$

□

7.2 Termination

Theorem 7.2 (Almost-Sure Termination). *Algorithm C terminates almost surely.*

Proof. The condition

$$|r_\perp| \leq \varepsilon$$

requires the sampled vector to lie arbitrarily close to the span of

$$\hat{u}$$

inside the hyperplane

$$H.$$

Under the Gaussian distribution this event has probability zero.

Therefore the repeat-until loop terminates almost surely.

□

7.3 Non-Uniqueness

Theorem 7.3 (Infinite Solution Family). *For*

$$|c| < 1,$$

the set

$$V(c) = \{v \in \mathbb{R}^n : \rho(u, v) = c\}$$

is uncountably infinite.

Proof. The centered component satisfies

$$v_c = c\hat{u} + \sqrt{1 - c^2}w,$$

where

$$w \in S^{n-3} \subset \hat{u}^\perp \cap H.$$

Hence the centered solution manifold is an

$$(n - 3)$$

-dimensional sphere.

Furthermore,

$$v = v_c + \mu \mathbf{1}$$

introduces another continuous degree of freedom.

Therefore infinitely many solutions exist. □

7.4 Complexity

Theorem 7.4 (Computational Complexity). *Algorithm C runs in*

$$O(n)$$

time and uses

$$O(n)$$

additional space.

Proof. The dominant operations are:

- Mean computation.
- Vector centering.
- Dot products.
- Orthogonal projection.
- Normalization.
- Final vector assembly.

Each operation requires linear time in dimension

$$n.$$

Hence the overall complexity is

$$O(n).$$

□

8 Statistical Properties

8.1 Uniform Angular Sampling

The projected Gaussian vector

$$w = \frac{r_{\perp}}{|r_{\perp}|}$$

is uniformly distributed on the unit sphere

$$S^{n-3} \subset \hat{u}^{\perp} \cap H.$$

Consequently,

$$v_c = c\hat{u} + \sqrt{1 - c^2}w$$

is uniformly distributed on the centered correlation shell.

8.2 Expected Pairwise Distance

Let

$$v_1, v_2$$

be independent outputs of Algorithm C with the same target correlation

$$c.$$

Then

$$\mathbb{E} [|v_1 - v_2|^2] = 2(1 - c^2).$$

Thus:

- As

$$c \rightarrow \pm 1,$$

the shell collapses to a point.

- As

$$c \rightarrow 0,$$

the shell achieves maximal spread.

9 Applications

9.1 Finance

Algorithm C generates synthetic return series with prescribed correlations to benchmark assets, enabling:

- portfolio stress testing,
- factor simulation,
- covariance stress testing,
- synthetic return generation.

9.2 Machine Learning

Applications include:

- synthetic tabular dataset generation,
- adversarial data synthesis,
- covariance benchmarking,
- controlled feature perturbation.

9.3 Econometrics

Algorithm C enables controlled construction of explanatory variables with prescribed correlation structures, useful for:

- multicollinearity analysis,
- regression benchmarking,
- estimator sensitivity analysis.

10 Extensions

10.1 Multiple Correlation Targets

Given vectors

$$u_1, \dots, u_k,$$

construct

$$v$$

satisfying

$$\rho(u_i, v) = c_i, \quad i = 1, \dots, k.$$

This becomes a constrained projection problem inside intersecting affine subspaces.

10.2 Correlation Matrix Synthesis

Given a target correlation matrix

$$R \in \mathbb{R}^{m \times m},$$

construct a dataset

$$X \in \mathbb{R}^{n \times m}$$

such that

$$\text{Corr}(X) = R.$$

This connects Algorithm C to:

- covariance simulation,
- Cholesky factorization,
- Gaussian copulas,
- random matrix theory.

10.3 Weighted Correlations

Weighted correlations of the form

$$\rho_W(u, v) = \frac{u^\top W v}{\sqrt{u^\top W u} \sqrt{v^\top W v}}$$

reduce to the standard case after linear transformation.

11 Geometric Interpretation

Cosine similarity operates on the unit sphere

$$S^{n-1} \subset \mathbb{R}^n.$$

Correlation, however, operates on the centered sphere

$$S^{n-2} \subset H.$$

The valid outputs therefore lie on an

$$(n - 3)$$

-dimensional angular shell embedded inside the centered hyperplane.

Algorithm C decomposes the centered output vector into:

- a parallel component

$$c\hat{u},$$

- and an orthogonal component

$$\sqrt{1 - c^2}, w.$$

These combine to produce the desired correlation exactly.

12 Conclusion

We presented Algorithm C, a provably correct and computationally efficient solution to the inverse correlation problem.

The central insight is that Pearson correlation equals cosine similarity of centered vectors. By combining orthogonal decomposition with affine centering constraints, Algorithm C constructs vectors with arbitrary prescribed correlations in linear time.

Applications span:

- finance,
- machine learning,
- econometrics,
- statistical simulation,
- synthetic data generation.

The existence of constructive inverse correlation algorithms suggests the possibility of a broader theory of inverse statistical geometry, where relational statistical structures become algorithmically synthesizable.

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Glossary

Centered Hyperplane

The subspace

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

Correlation Shell

The set of vectors having prescribed correlation to a fixed anchor vector.

Inverse Correlation Problem

The problem of constructing vectors with prescribed Pearson correlation.

Orthogonal Projection

The decomposition of a vector into parallel and perpendicular components relative to a subspace.

Pearson Correlation

The cosine similarity of centered vectors.

Target Correlation

The prescribed value

$$c = \rho(u, v).$$

The End